

The remainder of the paper is organized as follows. Section 2 introduces the model. Section 3 characterizes the equilibrium. Section 4 reviews the data and we calibrate the model to match salient empirical moments in section 5. Section 6 concludes.

2 Model

The model is populated by a constant measure M of workers who are employed in one of the I sectors of the economy. Each individual is born with an exogenous I -dimensional vector of skills $\vec{a}$, where each element characterizes her innate ability in any one of the $i \in \{1, \dots, I\}$ occupations in the economy, each corresponding to one of the sectors or industries. This vector of abilities is drawn from a known joint distribution $F(\vec{a})$. Moreover, the measure of workers is partitioned into G different groups, which are indexed by $g \in \{1, \dots, G\}$. Workers are born into one of these groups and membership is permanent. Moreover, we assume that the distribution of abilities is identical across groups, i.e., $F^g(\vec{a}) = F^{g'}(\vec{a}) = F(\vec{a})$.

Without loss of generality, occupation 1 is *teaching*. In this economy, teachers are distinct from other occupations since their technology produces human capital rather than the homogeneous final good in the remaining $I - 1$ sectors. This kind of distinction is salient since we are assuming that the economy is populated by overlapping generations of workers who accumulate human capital when young. When they reach working age (i.e., they are “old”), their human capital contributes toward the efficiency units of labor they supply to the labor market, including home production.

2.1 Preferences and Technologies

As in Roy (1951), workers make occupational choices based on their full range of idiosyncratic abilities $\vec{a}$ and the corresponding “payoffs.” To this basic mechanism we add Hsieh et al. (2019)-style forces that distort the allocation of skill across occupations and a human capital formation technology with time, good, and teacher inputs.

The timing convention is similar to Hsieh et al. (2019). In the first period, prospective workers – or students – make occupation-specific human capital investments. Since there is no uncertainty, students have perfect foresight and their investment decision will depend on their fully anticipated occupational choice in period 2. They retire at the end of their working period and are replaced by a new cohort of students with measure $\frac{M}{2}$. The environment is stationary in the sense that the members of each cohort draw their ability vectors from the same distribution F_a .

The worker’s utility depends on her consumption and time devoted to human capital invest-

ment:

$$U = \max_{C'_g, \{s_{i,g}, e_{i,g}\}_{i=1}^I} \left\{ \mu \ln C'_g + \ln(1 - s_{i,g}) \right\}_{i=1}^I \quad (1)$$

where

$$C'_g = (1 - t')(1 - \tau_{i,g}^{\omega'}) \omega'_{i,g}(h_{i,g}) - \sum_{i=1}^I (1 + \tau_{i,g}^e) e_{i,g} \quad (2)$$

$$h'_{i,g} = \kappa(h_{T,g}, a_i) (s_{i,g})^\phi (e_{i,g})^\eta (N(h_{T,g}))^{-\sigma} \quad (3)$$

$$\kappa(h_{T,g}, a_i) = (h_{T,g})^\beta (a_i)^\alpha. \quad (4)$$

C'_g is consumption of the homogeneous final good when old, $s_{i,g}$ is the time allocated to occupation-specific human capital formation when young, and $e_{i,g}$ are units of the final good invested in occupation-specific human capital formation when workers are still students. These investments and the corresponding consumption in the following period are occupation-specific and indexed by the subscript i .⁴ The periodic time endowment is set to unity. $1 - s_{i,g}$ is leisure time when young, and labor is supplied inelastically when individuals reach working age.

Labor income is taxed at a constant marginal rate $t \in [0, 1)$ and $\tau_{i,g}^\omega < 1$ is a group- g -and-occupation- i -specific labor market distortion. We allow for the possibility of *negative* distortions. Finally, $\tau_{i,g}^e$ is a group and occupation-specific educational barrier that raises the cost of good investments in the human capital accumulation technology.

We assume that the draws from $F_a(\cdot)$ are i.i.d. across agents but we allow for the possibility that draws are not independent across occupations.

The production technologies in occupations $2, \dots, I$ are linear in efficiency units of labor $h_{i,g}$ with productivities $\{A_i\}_{i=2}^I$:

$$y_{i,g} = A_i \times h_{i,g}. \quad (5)$$

Since human capital is the only factor of production, we can characterize the income of a worker who is employed in a non-teaching job by:

$$\omega_i(h_{i,g}) = y_{i,g}. \quad (6)$$

Teaching is distinct from other occupations since it “transforms” students with skill endowment $\vec{a}$ into workers with human capital $\vec{h}$ based on equation (3). The acquisition of human capital depends on the teacher’s input $h_{T,g}$, the student’s talent $\vec{a}$, the student’s time and

⁴Nothing prevents students from making investments in human capital in multiple occupations. Since there is no uncertainty in future payoffs, they will, however, select only one in equilibrium.

good investments $s_{i,g}$ and $e_{i,g}$, as well as class size $N(h_{T,g})$, which we allow to vary with the teacher's $h_{T,g}$. Note that students already select an occupation at this stage and the optimal investment depends on a single element, say a_i , of $\vec{a}$.

2.2 Occupational Choice

Individuals will choose the occupation that delivers the highest lifetime utility. The workers' choice between non-teaching occupations is typical in the sense that it is based on a "conventional" Roy (1951)-type model that has been used extensively in the previous literature. Teaching, on the other hand, is modeled in a way that deviates from the standard assumptions in two key respects: (1) the teachers' human capital is an input into the *students'* human capital production function rather than the production function for final goods and (2) the teacher's wage $\omega_{T,g}(h_{T,g})$ need not be linear in $h_{T,g}$. This implies that the choice of becoming a teacher doesn't just depend on a worker's comparative advantage, it also depends on her *absolute* advantage.

Due to the special nature of teaching, we will discuss the key characteristics based on a version of the model with $I = 2$. This allows us to highlight the novel elements of the model more effectively. In this version, we label the two occupations with subscripts T (for *teaching*) and O (for *other*). Later, and especially when we solve the quantitative version of the model in order to match salient moments in U.S. data, we will re-introduce the additional $I - 2$ occupations.

2.2.1 Teachers and Class Size

In equations (3) and (4), the student's ability in occupation i , denoted a_i , and the teacher's human capital h_T are complements. This implies positive sorting between teachers and students. This effect, however, is tempered by the teacher's span-of-control over the students in her class. $N(\cdot)$ denotes the class size and the coefficient $-\sigma$ captures the stylized fact that in larger classes teachers can pay less attention to one individual student. Here, we allow class size to depend on $h_{T,g}$ and we are focusing on $N(h_{T,g})$ such that – for given $e_{i,g}$ and $s_{i,g}$ – students are indifferent between teachers with different levels of human capital, say $h_{T,g}$ and $h'_{T,g}$:

$$N(h'_{T,g}) = \left(\frac{h'_{T,g}}{h_{T,g}} \right)^{\frac{\beta}{\sigma}} N(h_{T,g}). \quad (7)$$

This implies that better teachers work in larger classrooms.

All $\frac{M}{2}$ students in the economy attend school. Therefore, the following resource constraint

must be satisfied:

$$\frac{M}{2} = \int_0^\infty N(h_{T,g}) d\mathcal{F}_T(h_{T,g}) \quad (8)$$

where $\mathcal{F}_T(\cdot)$ is the c.d.f. of human capital in teaching.

The resource constraint (8) anchors the class size distribution and equation (7) characterizes the size distribution of classes.

Proposition 1 (Class size distribution) *For a given teacher with human capital $h_{T,g}$, her class size is characterized by:*

$$\left(\frac{2\tilde{H}_T}{M} \right)^\sigma = h_{T,g}^\beta N(h_{T,g})^{-\sigma}, \quad (9)$$

where $\tilde{H}_T = \sum_{g=1}^G \int_0^\infty h_{T,g}^{\frac{\beta}{\sigma}} d\mathcal{F}_T(h_{T,g})$.

2.2.2 Student-Teacher Matching

While this guarantees that students are indifferent between different teachers, it may be the case that better teachers prefer to work with higher ability students. This can happen, for instance, if students make human capital investments in terms of goods as a function of the teacher's human capital $h_{T,g}$. The increased investment then increases the teacher's marginal product, which would be a force for positive sorting.

For simplicity, we assume that students make investments as a function of a and occupation i , but not $h_{T,g}$. The joint assumption that $e_{i,g}(a_i)$ is a function of a (but not $h_{T,g}$) and that $N(h_{T,g})$ is a function of $h_{T,g}$ (but not a_i) implies that the student-teacher matching is random, i.e., the human capital production function is modular in $h_{T,g}$ and a_i .⁵

2.3 Values

The structure we impose on class size and student-teacher matching simplifies the characterization of an equilibrium considerably. While idiosyncratic heterogeneity still matters, the aggregate state can be summarized by a single moment of the distribution of human capital among current teachers (which we discuss in more detail below) rather than the entire distri-

⁵The anecdotal evidence suggests that student-teacher assignments don't involve systematic sorting at the school level. There is, however, evidence in support of sorting between families with highly educated parents and high-ability teachers into better funded schools or school districts. Identifying the empirically relevant extent of this type of sorting is work in progress. Future versions of our model may eventually incorporate it.

bution of $\{h_{T,g}\}_{g=1}^G$. Thanks to this feature we can characterize stationary equilibria as well as full transition paths sharply.

The idiosyncratic state of a student in this model consists of the I -dimensional vector of abilities $\vec{a}$. Recall that we assume $I = 2$ for the time being, where T denotes teaching and O lumps together all other occupations. An individual teachers' human capital h_T is not part of the idiosyncratic state space. A single moment of the distribution off all teachers' human capital is part of the aggregate state space.

A student in group g solves a maximization problem which entails a discrete occupational choice as well as continuous human capital investments in terms of time and final goods. Formally, she solves:

$$V_g(a_T, a_O, \tilde{H}_T) = \max_{\{s_{O,g}, s_{T,g}, e_{O,g}, e_{T,g}\}} \left\{ V_{O,g}(a_O, \tilde{H}_T), V_{T,g}(a_T, \tilde{H}_T) \right\} \quad (10)$$

where

$$\begin{aligned} V_{O,g}(a_O, \tilde{H}_T) = & \ln \left(1 - s_{O,g}(a_O, \tilde{H}_T) \right) \\ & + \mu \ln \left[h'_{O,g} A'_O (1 - t') (1 - \tau_{O,g}^{\omega'}) \right. \\ & \left. - e_{O,g}(a_O, \tilde{H}_T) (1 + \tau_{O,g}^e) \right], \end{aligned} \quad (11)$$

$$\begin{aligned} V_{T,g}(a_T, \tilde{H}_T) = & \ln \left(1 - s_{T,g}(a_T, \tilde{H}_T) \right) \\ & + \mu \ln \left[\omega' (h'_{T,g}, \tilde{H}'_T) (1 - t') (1 - \tau_{T,g}^{\omega'}) \right. \\ & \left. - e_{T,g}(a_T, \tilde{H}_T) (1 + \tau_{T,g}^{e'}) \right] \end{aligned} \quad (12)$$

To make further progress, we need to characterize an occupational choice threshold along which a student with abilities (a^T, a^O) is indifferent between teaching and production work. More formally, the threshold is a function

$$a_{T,g}^*(a_O) = \bar{a}_g(a_O, \tilde{H}_T)$$

such that

$$V_{O,g}(a_O, \tilde{H}_T) = V_{T,g}(a_{T,g}^*(a_O), \tilde{H}_T), \text{ for all } a_O \in (0, \infty) \quad (13)$$

Unless it's necessary, we will omit the aggregate state argument $\tilde{H}_T$ from $\bar{a}_g(\cdot)$ in order to keep the notation clean and tidy.⁶

⁶We will show later that the occupational threshold is indeed independent from $\tilde{H}_T$. At this point, this result is not (yet) obvious.

The optimization problem is subject to the following constraints:

$$t \left[\sum_{g=1}^G \int_0^\infty \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da + \sum_{g=1}^G \int_0^\infty \omega_{O,g}(h_{O,g}(a)) f_{O,g}(a) da \right] = \sum_{g=1}^G \int_0^\infty \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da \quad (14)$$

where

$$f_{T,g}(a) = \int_0^{\bar{a}_g^{-1}(a)} f(a, b) db, \quad (15)$$

$$f_{O,g}(b) = \int_0^{\bar{a}_g(b)} f(a, b) da. \quad (16)$$

$f(a, b)$ is the p.d.f. associated with the c.d.f. $F(\cdot)$ from section 2.1 in the special case with $I = 2$, $f_{T,g}(a)$ describes today's distribution of teachers, and $f_{O,g}(a)$ characterizes the distribution of other workers. Workers have perfect foresight and correctly anticipate next period's tax rate t' and future labor market distortions $\left\{ \tau_{O,g}'^{\omega}, \tau_{T,g}'^{\omega} \right\}_{g=1}^G$ that satisfy (14)-(16) one period from today.⁷

The aggregate laws of motion for human capital in teaching ($\tilde{H}_T$) and "other" (H_O) are:

$$H_O' = \sum_{g=1}^G \int_0^\infty \left(\frac{2\tilde{H}_T}{M} \right)^\sigma a^\alpha s_{O,g} \left(a, \tilde{H}_T \right)^\phi e_{O,g}(a, \tilde{H}_T)^\eta f_{O,g}(a) da \quad (19)$$

$$\tilde{H}_T' = \sum_{g=1}^G \int_0^\infty \left(\left(\frac{2\tilde{H}_T}{M} \right)^\sigma a^\alpha s_{T,g} \left(a, \tilde{H}_T \right)^\phi e_{T,g}(a, \tilde{H}_T)^\eta \right)^{\frac{\beta}{\sigma}} f_{T,g}(a) da \quad (20)$$

Teachers are public sector employees and their compensation is funded by tax revenues. Equation (14) is the (local) government's balanced budget requirement, which is a salient feature in the U.S. context.⁸

⁷In the more general case with $I > 2$, $i = 1$ indexes teachers and the measures of teachers and other workers from equations (15) and (16) are given by:

$$f_{T,g}(a_T) = \sum_{i=2}^I \int_0^{\bar{a}_{i,g}^{-1}(a_T)} \Pi_{j \neq i} F \left(\left[\frac{A_i}{A_j} \cdot \frac{1 - \tau_{i,g}'^{\omega}}{1 - \tau_{j,g}'^{\omega}} \cdot \left(\frac{1 + \tau_{i,g}^e}{1 + \tau_{j,g}^e} \right)^{-\eta} \right]^{\frac{1}{\alpha}} \cdot a_i \right) f_{i,g}(a) da \text{ and} \quad (17)$$

$$f_{i,g}(a_i) = F(\bar{a}_{i,g}(a_i)) \Pi_{j \neq i} F \left(\left[\frac{A_i}{A_j} \cdot \frac{1 - \tau_{i,g}'^{\omega}}{1 - \tau_{j,g}'^{\omega}} \cdot \left(\frac{1 + \tau_{i,g}^e}{1 + \tau_{j,g}^e} \right)^{-\eta} \right]^{\frac{1}{\alpha}} \cdot a_i \right), \text{ respectively.} \quad (18)$$

⁸The current version of the model has a single location. Arguably, a more salient environment would feature multiple locations, each of which is subject to a local balanced budget requirement. This extension introduces

Equation (19) is the aggregation over h_O of the current cohort of students, while equation (20) is the $\left(\frac{\beta}{\sigma}\right)^{\text{th}}$ moment of the distribution of h_T of the current cohort of students.

Note that H'_O depends on $\tilde{H}'_T$ but not on H_O . This reflects the assumption that the acquisition of human capital by today's students – regardless of the occupational choice – depends on the quality of today's teachers but not on the human capital of today's production workers.

The teachers' wage profile ω_T is exogenous. We assume that ω_T is strictly increasing in the teacher's human capital h_T and continuously differentiable. We allow, in particular, for the possibility that ω_T is not linear in h_T .

3 Equilibrium

Definition 1 *Given next period's tax rate t' , the labor market distortions and educational barriers $\left\{ \left\{ \tau_{i,g}^e, \tau_{i,g}^{\omega'} \right\}_{i=1}^I \right\}_{g=1}^G$, the occupational choices of today's old agents and aggregate human capital $\tilde{H}_T$ and H_O , the equilibrium consists of the individual investment decisions of young agents, $\{e_{T,g}, s_{T,g}, e_{O,g}, s_{O,g}\}$, the occupational choice boundary $a_{T,g}^*(a_O)$, the corresponding densities $f_{T,g}(\cdot)$ and $f_{O,g}(\cdot)$, and occupation- and group-specific wage profiles $\{\omega_{T,g}, \omega_{O,g}\}$ such that:*

1. *Given the wage profiles, each individual's choice of goods investment $\{e_{T,g}, e_{O,g}\}$ and time investment $\{s_{T,g}, s_{O,g}\}$ maximizes (10).*
2. *The aggregate human capital stocks follow the laws of motion given by (19) and (20).*
3. *The government's budget constraint (14) is satisfied.*

3.1 Human Capital Investment

Let's first characterize the optimal human capital investment for a prospective teacher with ability a_T from group g .

The F.O.C.s for $s_{T,g}$ and $e_{T,g}$, respectively, after a few steps of algebra are:

$$(1 + \tau_{T,g}^e) e_{T,g} \left(\frac{\sigma}{\eta\beta} - 1 \right) = (1 - s_{T,g}) \mu (1 - t') (1 - \tau_{T,g}^{\omega'}) \cdot \frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \cdot \frac{\partial h'_{T,g}}{\partial s_{T,g}} \quad (21)$$

$$1 + \tau_{T,g}^e = (1 - t') (1 - \tau_{T,g}^{\omega'}) \cdot \frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \cdot \frac{\partial h'_{T,g}}{\partial e_{T,g}}, \quad (22)$$

additional complexities and we focus on this simple version for now.

Let $\tau_{T,g} \equiv \frac{(1-t') (1-\tau_{T,g}^{\omega'})}{1+\tau_{T,g}^e}$ be a composite wedge that takes into account the rate at which future income is taxed (t'), the extent of labor market discrimination ($\tau_{T,g}^{\omega'}$), and educational barriers ($\tau_{T,g}^e$). Note that $\tau_{T,g} = 1$ when $t' = \tau_{T,g}^{\omega'} = \tau_{T,g}^e = 0$.

The amount of time and goods this individual invests in human capital is then given by:

$$s_{T,g} = \frac{\mu\phi}{\mu\phi + \eta \cdot \tau_{T,g} \cdot \frac{\omega'_{T,g}(h'_{T,g})}{e_{T,g}} - \eta} \quad (23)$$

$$e_{T,g} = \left(\tau_{T,g} \cdot \eta \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \cdot \left(\frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \right) \cdot s_{T,g}^\phi \cdot a_T^\alpha \right)^{\frac{1}{1-\eta}} \quad (24)$$

The F.O.C.s for prospective production workers (i.e., those labeled “other”) are analogues of (23) and (24) with the simplification that the wage profile in production (as opposed to teaching) is linear in $h_{O,g}$ and the optimal amount of time and resources invested in human capital is given by:

$$s_{O,g} = \frac{\mu\phi}{\mu\phi + 1 - \eta} \quad (25)$$

$$e_{O,g} = \left(\tau_{O,g} \cdot \eta \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \cdot A'_O \cdot s_{O,g}^\phi \cdot a_O^\alpha \right)^{\frac{1}{1-\eta}} \quad (26)$$

where, analogously to the problem of a prospective teacher, we define $\tau_{O,g} \equiv \frac{(1-t') (1-\tau_{O,g}^{\omega'})}{1+\tau_{O,g}^e}$.

3.2 The Aggregate Laws of Motion

After substituting (23)-(26) into (19) and (20) we can characterize the equilibrium laws of motion for $\tilde{H}_T$ and H_O by:

$$\tilde{H}'_T = \eta^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\beta}{1-\eta}} \cdot \sum_{g=1}^G \tau_{T,g}^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \int_0^\infty \left(\frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \right)^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot s_{T,g}^{\frac{\beta\phi}{\sigma(1-\eta)}} \cdot a^{\frac{\beta\alpha}{\sigma(1-\eta)}} f_{T,g}(a) da \quad (27)$$

and

$$H'_O = \eta^{\frac{\eta}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \sum_{g=1}^G \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot \int_0^\infty A'_O{}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da. \quad (28)$$

We can show that this economy has a steady state in $\tilde{H}_T$ and we can characterize it implicitly in a single equation:

$$\tilde{H}_T = \eta^{\frac{\beta\eta}{\sigma(1-\eta-\beta)}} \cdot \left(\frac{2}{M}\right)^{\frac{\beta}{1-\eta-\beta}} \cdot \left[\sum_{g=1}^G \tau_{T,g}^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \int_0^\infty \left(\frac{\partial\omega_{T,g}}{\partial h_{T,g}}\right)^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot s_{T,g}^{\frac{\beta\phi}{\sigma(1-\eta)}} \cdot a^{\frac{\beta\alpha}{\sigma(1-\eta)}} f_{T,g}(a) da \right]^{\frac{1-\eta}{1-\eta-\beta}} \quad (29)$$

3.3 Occupational Choice

Proposition 2 (Occupational Choice Boundary) *The occupational choice boundary, denoted $\bar{a}_{T,g}(a_O)$, depends on the magnitude of the educational barriers, the labor market frictions, and the student's teaching-specific human capital $h'_{T,g}$. It does not, however, depend on either current or future aggregate human capital in teaching $\tilde{H}_T$. It is characterized by:*

$$\left(\frac{a_T}{a_O}\right)^\alpha \cdot \frac{\tau_{T,g}}{\tau_{O,g}} \cdot \left(\frac{1 + \tau_{T,g}^e}{1 + \tau_{O,g}^e}\right)^{1-\eta} \cdot \left(\frac{1 - s_{T,g}}{1 - s_{O,g}}\right)^{\frac{1-\eta}{\mu}} \cdot \left(\frac{s_{T,g}}{s_{O,g}}\right)^\phi \cdot \frac{\partial\omega'_{T,g}}{\partial h'_{T,g}} \cdot \left(\frac{\frac{\omega'_{T,g}}{\partial\omega'_{T,g}} \cdot h'_{T,g} - \eta}{1 - \eta}\right)^{1-\eta} = 1. \quad (30)$$

4 Data

We parameterize the model with data from longitudinal surveys that cover American youth at three different times after World War II: Project TALENT, NLSY 79, and NLSY 97. While the two NSLY surveys have been used extensively in the literature, Project TALENT is somewhat less known and we describe it in some detail below.

4.1 Project TALENT

Project TALENT surveyed approximately 377,000 students enrolled in high school in 1960. The vast majority of these students were in the 14-18 year age range with a few outliers outside this bracket (on both sides). The sample of surveyed students was representative and accounted for 5 percent of the high school population that year. The project was funded by the United States Office of Education (the precursor of the Department of Education) and its goal was to collect high quality data on the achievements, aptitudes, and interests of high